

Program 10: Demonstrate kubernetes dashboard and CLI for cluster monitoring

Step 1: Create deployment.yaml file

```
apiVersion: v1
kind: Deployment
metadata:
  name: nginx-deploy
  labels:
    app: nginx
spec:
  replicas: 2
  selector:
    matchLabels:
      app: nginx
  template:
    metadata:
      labels:
        app: nginx
    spec:
      containers:
        - name: nginx
          image: nginx:latest
          ports:
            - containerPort: 80
```

Step 2: Create service.yaml file

```
apiVersion: v1
kind: Service
metadata:
  name: nginx-service
spec:
  selector:
    app: nginx
  ports:
    - protocol: TCP
      port: 80
      targetPort: 80
  type: NodePort
```

Step 3: Start minikube

```
1rv24mc040-haripriya@haripriya-ASUS:~/Devops/program10$ minikube start --cpus=2 --memory=4096
🐳 minikube v1.37.0 on Ubuntu 24.04
🔧 Using the docker driver based on existing profile
❗ You cannot change the memory size for an existing minikube cluster. Please first delete the cluster.
👉 Starting "minikube" primary control-plane node in "minikube" cluster
📦 Pulling base image v0.0.48 ...
🔄 Updating the running docker "minikube" container ...
❗ Failing to connect to https://registry.k8s.io/ from inside the minikube container
💡 To pull new external images, you may need to configure a proxy: https://minikube.sigs.k8s.io/docs/reference/networking/proxy/
🔧 Preparing Kubernetes v1.34.0 on Docker 28.4.0 ...
🔍 Verifying Kubernetes components...
   ▪ Using image gcr.io/k8s-minikube/storage-provisioner:v5
🌟 Enabled addons: storage-provisioner, default-storageclass

❗ /usr/local/bin/kubectl is version 1.31.0, which may have incompatibilities with Kubernetes 1.34.0.
   ▪ Want kubectl v1.34.0? Try 'minikube kubectl -- get pods -A'
🏠 Done! kubectl is now configured to use "minikube" cluster and "default" namespace by default
```

Step 4: Apply the yaml files

```
1rv24mc040-haripriya@haripriya-ASUS:~/Devops/program10$ kubectl apply -f deployment.yaml
deployment.apps/nginx-deploy created

1rv24mc040-haripriya@haripriya-ASUS:~/Devops/program10$ kubectl apply -f service.yaml
service/nginx-service created
```

Step 5: Check the resources

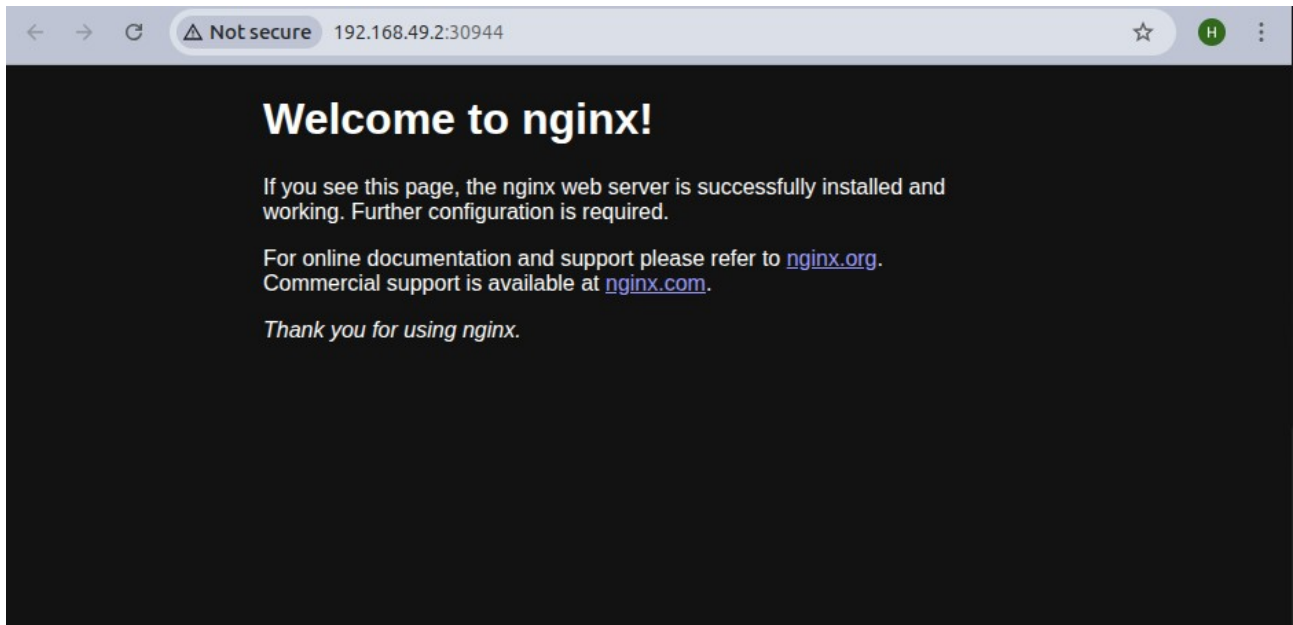
```
1rv24mc040-haripriya@haripriya-ASUS:~/Devops/program10$ kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
kubia                               1/1     Running   4 (8m32s ago)    10d
nginx-deploy-6f9664446b-9fvk9       1/1     Running   0           3m
nginx-deploy-6f9664446b-fj7ps       1/1     Running   0           3m
nginx-deployment-6f9664446b-7bzg4   1/1     Running   3 (8m42s ago)    7d
nginx-deployment-6f9664446b-r695t   1/1     Running   3 (8m42s ago)    7d
nodejs-pod                          1/1     Running   3 (8m32s ago)    7d
```

Step 7: Access the application

```
1rv24mc040-haripriya@haripriya-ASUS:~/Devops/program10$ minikube service nginx-service
```

NAMESPACE	NAME	TARGET PORT	URL
default	nginx-service	80	http://192.168.49.2:30944

```
🌐 Opening service default/nginx-service in default browser...
```



```
1rv24mc040-haripriya@haripriya-ASUS:~/Devops/program9$ minikube dashboard
```

```
🐼 Verifying dashboard health ...
```

```
🚀 Launching proxy ...
```

```
🐼 Verifying proxy health ...
```

```
🎉 Opening http://127.0.0.1:33337/api/v1/namespaces/kubernetes-dashboard/services/http:kubernetes-dashboard:/proxy/ in your default browser...
```

```
Opening in existing browser session.
```

Workload Status

Running: 2 Deployments Running: 6 Pods Running: 2 Replica Sets

Deployments

Name	Images	Labels	Pods	Created
nginx-deploy	nginx:latest	app: nginx	2 / 2	16 minutes ago
nginx-deployment	nginx:latest	-	2 / 2	7 days ago

Pods

Name	Images	Labels	Node	Status	Restarts	CPU Usage (cores)	Memory Usage (bytes)	Created
nginx-deploy-6f9664446b-9fvk9	nginx:latest	app: nginx pod-template-hash: 6f9664446b	minikube	Running	0	0.00m	10.92Mi	16 minutes ago
nginx-deploy-6f9664446b-67ps	nginx:latest	app: nginx pod-template-hash: 6f9664446b	minikube	Running	0	0.00m	10.58Mi	16 minutes ago